

2025 AMC 10B Problem 15 - Solution

Review the full statement and step-by-step solution for 2025 AMC 10B Problem 15. Great practice for AMC 10, AMC 12, AIME, and other math contests

2025 AMC 10B Problem 15



Problem:

The sum $\sum_{k=1}^{\infty} \frac{1}{k^3 + 6k^2 + 8k}$ can be expressed as $\frac{a}{b}$, where a and b are relatively prime positive integers. What is $a + b$?

Answer Choices:

- A. 89
- B. 97
- C. 102
- D. 107
- E. 129



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Solution:

By partial fraction decomposition, we have $\frac{1}{k^3 + 6k^2 + 8k} = \frac{1}{8} \left(\frac{1}{k} - \frac{2}{k+2} + \frac{1}{k+4} \right)$, so we may write:

$$\begin{aligned}
 8 \sum_{k=1}^{\infty} \frac{1}{k^3 + 6k^2 + 8k} &= \sum_{k=1}^{\infty} \left(\frac{1}{k} - \frac{2}{k+2} + \frac{1}{k+4} \right) \\
 &= \sum_{k=1}^{\infty} \left(\frac{1}{k} - \frac{1}{k+2} \right) + \sum_{k=1}^{\infty} \left(\frac{1}{k+4} - \frac{1}{k+2} \right) \\
 &= \sum_{k=1}^{\infty} \left(\frac{1}{k} - \frac{1}{k+2} \right) - \sum_{k=1}^{\infty} \left(\frac{1}{k+2} - \frac{1}{k+4} \right) \\
 &= \sum_{k=1}^{\infty} \left(\frac{1}{k} - \frac{1}{k+2} \right) - \sum_{k=3}^{\infty} \left(\frac{1}{k} - \frac{1}{k+2} \right) \\
 &= \sum_{k=1}^2 \left(\frac{1}{k} - \frac{1}{k+2} \right)
 \end{aligned}$$

Alternatively, we may note the individual sums themselves telescope and compute those sums first. Either method gives an answer of $\frac{1}{8} \left(\frac{1}{1} + \frac{1}{2} - \frac{1}{3} - \frac{1}{4} \right) = \frac{1}{8} \cdot \frac{11}{12} = \frac{11}{96}$ which gives a sum of $11 + 96 = \boxed{\text{(D) } 107}$.

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